IAEA nuclear fusion

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To cite this article: B. Lehnert *et al* 1966 *Nucl. Fusion* **6** 231

CRITICAL VOLTAGE OF A ROTATING PLASMA

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The accessible voltage levels have been determined for a rotating plasma device in which the confinement region surrounds a ring-shaped coil that is suspended by a single rod. (i) There exists a sharply defined critical voltage and a corresponding velocity limit which cannot be exceeded by a moderately large power input. This is clearly shown with the present apparatus where the plasma voltage slowly approaches the critical level from below and where the insulator consists only of a narrow rod. (ii) The critical voltage increases with the mean radial ratio r_0/r_w in the plasma. This ratio is defined by a magnetic field line which intersects the midplane of the device at a radial distance r_0 from the axis of symmetry, and cuts the end insulator at the corresponding distance r_w. (iii) The results can be explained by an interaction between the plasma and a neutral gas layer at the insulator surface that limits the velocity to the critical value v_c earlier introduced by Alfvén. As a consequence, the velocity limit becomes $(r_0/r_w)v_c$ in the equatorial plane. Deductions of the corresponding voltage limit agree with the experiments. (iv) The plasma is heavily disturbed even by a very small obstacle which is introduced in the equatorial plane. This indicates that the losses of the system are small under normal operation. (v) So far the stability is not influenced by the fact that most of the field lines pass freely around the confinement region and do not end upon an insulator surface.

1 Introduction

In 1954 Alfvén [1] suggested that there exists an enhanced ionization process between a charged and a neutral gas cloud. It was postulated to take place as soon as the relative velocity between the clouds approaches the critical value $v_c = (2\,e\,\Phi_i/m_i)^{\frac{1}{2}}$, where Φ_i is the ionization potential and m_i the ion mass. This suggestion has been supported by a number of experiments, both with homopolar devices [2, 3, 4] and with coaxial plasma guns [5, 6]. Within a wide range of the experimental parameters, the observed plasma voltage was limited to a value close to that calculated from v_c. Several attempts have been made to explain these results theoretically [7, 8, 9, 10, 11], but the discussions has not yet been settled.

There further exist rotating plasma devices in which there is a voltage limit which definitely exceeds the value calculated from v_c in the equatorial midplane. Examples of this are given by the Ixion [12] and by a device earlier described by us (henceforth called FI) [13, 14, 15].

In all rotating plasma experiments, an interaction between the plasma and the end insulators is probably responsible for the voltage limitation [15, 16, 17]. Ions and electrons will escape along the magnetic field lines to the insulators, where a recombination takes place to produce a backflux of neutrals which form a narrow layer of neutral gas close to the insulator surface. When the velocity of rotation approaches v_c in the layer, a violent interaction starts between the plasma and the neutral gas. Then, a further input of energy will damage the insulator and produce a high gassing rate, but lead to no further increase in the plasma velocity. Consequently, the iso-rotation law will limit the velocity in the equatorial plane to $(r_0/r_w)v_c$, where r_0 is the radial distance of a field line which intersects the insulator surface at r_w. For devices with large radial ratios r_0/r_w, such as the Ixion and FI, we should therefore expect the voltage limit to be higher than for a Homopolar where $r_0/r_w \sim 1$.

This explanation is supported by the fact that the voltage levels given by v_c can be recovered in FI if an insulating plate is placed near the equatorial plane [13]. Further, in Homopolar V there does not seem to exist any voltage limit as long as the plasma does not touch the end insulators [18]. Finally, during the start of a rotating plasma the velocity is limited to v_c in the equatorial plane because there is neutral gas within the entire plasma volume [19, 20].

In this paper further proofs will be given that there exists a sharply defined critical voltage which depends on the radial ratio. For this purpose a device has been used in which the critical voltage can be approached slowly from below and in which the radial ratio can be varied continuously.

2 Experimental arrangement

The discharge chamber of the present device (henceforth called FII) is shown in fig. 1. The magnetic field is generated by a ring-shaped main coil and two auxiliary coils. A calculation of the field pattern has been made with a computer [21]. The main coil is suspended on a spoke-shaped rod which is fixed at the inside of the main coil. The rod is surrounded by an insulator made of aluminium ceramics. The field lines pass freely around the main coil, except for an azimuthal angle of about 40 degrees were they end on the insulator. The magnetization current I_m of the coil system has been varied up to 3.4 kA, which corresponds to a mean field strength B_0 in the equatorial plane of about 0.6 V sec/m^2.

The discharge is run between two anode rings and a movable cathode plate that are all made of molybdenum. The position of the electrodes and the magnetic field lines determine the boundaries of the plasma confinement region that intersect the equatorial plane at r_{01} and r_{02}.

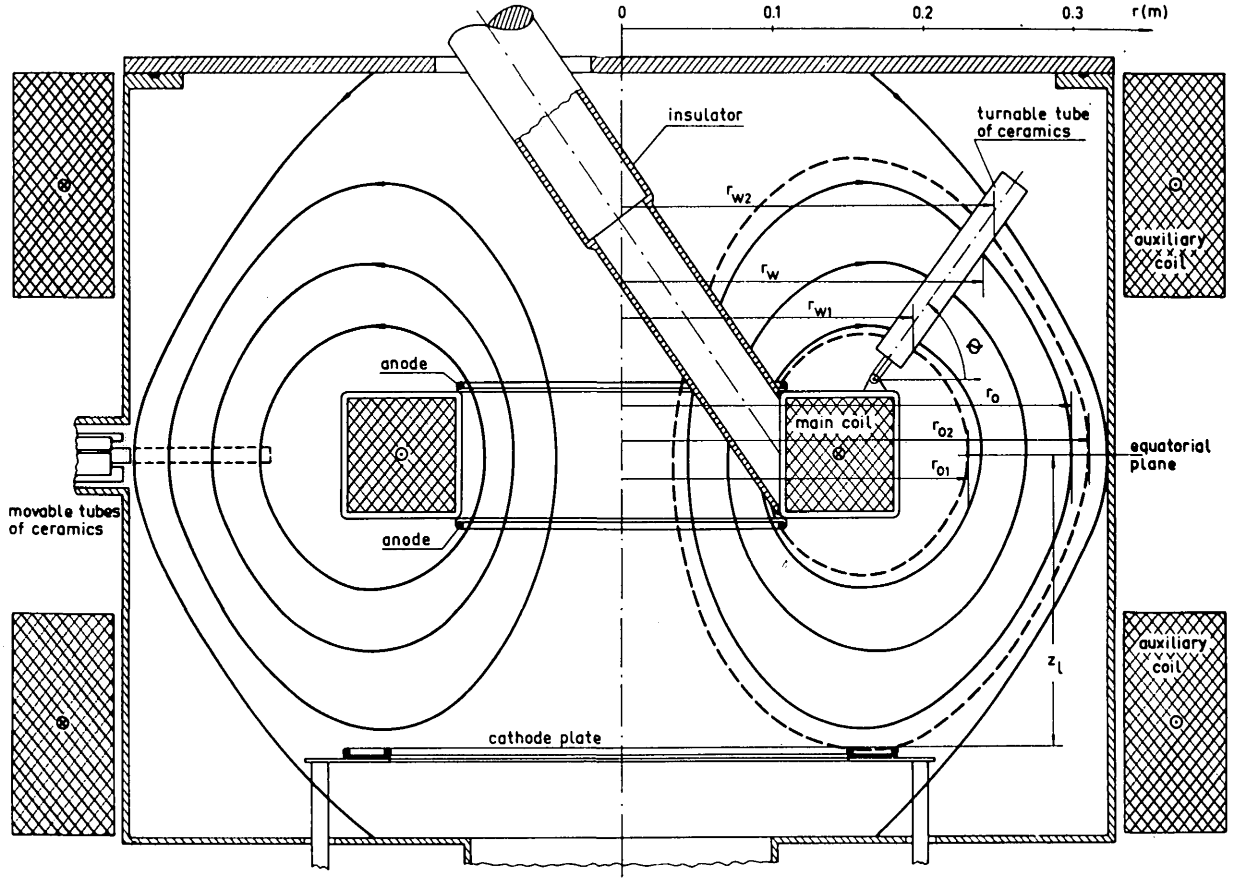

Fig. 1 The discharge chamber with the magnetic field configuration of device F II. The discharge takes place between two anode rings and a cathode plate. The confinement region is situated between the dashed lines in the right hand part of the figure.

In the experiments described in sect. 3B, a turnable ceramic tube, 21 mm in diameter, has been inserted across the confinement region. By this arrangement, the radial ratio r_0/r_w can be varied continuously in the plane through the tube and the axis of symmetry. Since the plasma is revolving rapidly around the axis, we shall later see from the experiments that this has the same effect on the voltage limit as a conical wall which extends itself around the whole perimeter of the device.

In the left-hand part of fig. 1 there are further a bundle of ceramic tubes that have diameters of 1.6, 5, 8, 10, 16, 20, and 24 mm. Each of these tubes can be moved into the equatorial region in order to study

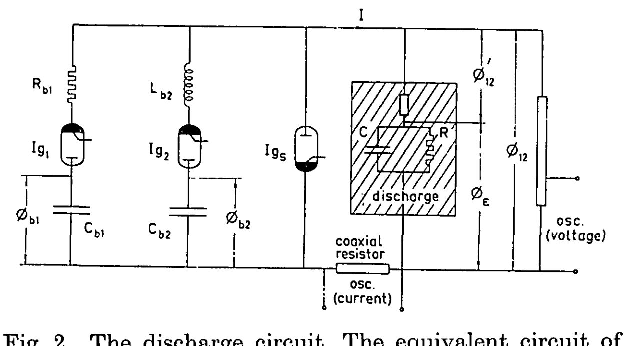

Fig. 2 The discharge circuit. The equivalent circuit of the plasma is indicated within the shaded area.

the corresponding influence on the plasma losses and the voltage levels, as described in sect. 3C.

The discharges have been performed with the cathode plate at a distance $z_l = 0.187$ m from the equatorial plane and with $r_{01} = 0.235$ m and $r_{02} = 0.305$ m. All experiments have been run with hydrogen, except in sect. 3A where deuterium also has been used.

Fig. 2 shows the discharge circuit. The discharge is started by connecting the condenser bank C_{b1} with the electrodes. After a fully ionized and rather slowly rotating plasma has been created, the second bank is connected at time t_2 through the inductance L_{b2}. The latter limits the current growth and produces a slow acceleration by which the plasma velocity gradually approaches the critical velocity level from below. Finally, at time t_s the stored angular momentum is controlled by means of a short-circuit. All experiments have been carried out with $C_{b1} = 82.4\ \mu$F, $C_{b2} = 144\ \mu$F and $R_{b1} = 0.985\ \Omega$. The inductance is $L_{b2} = 441.5\ \mu$H in fig. 4 and $L_{b2} = 436\ \mu$H in all other cases.

3 Experimental results

3A The Critical Voltage

When the plasma is accelerated by the second bank with comparatively small currents I, the voltage

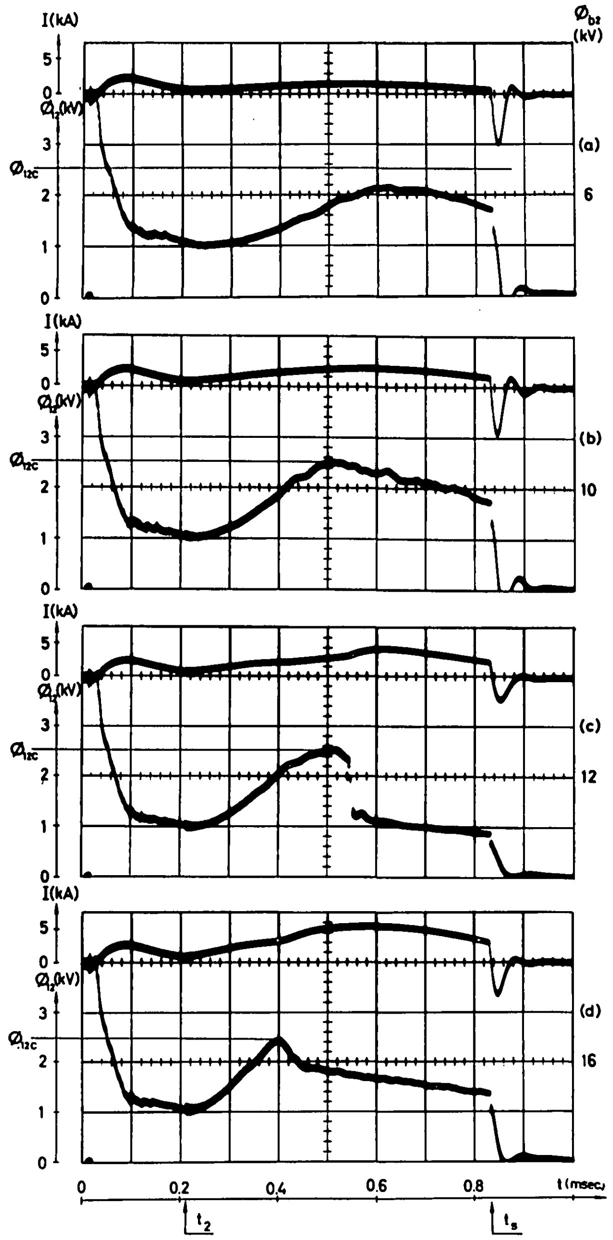

Fig. 3 Plasma current I and voltage Φ_{12} at increasing voltage Φ_{b2} of second bank: (a) 6 kV; (b) 10 kV; (c) 12 kV; (d) 16 kV. $I_m = 2.45$ kA, $p_0 = 34$ mTorr., $\Phi_{b1} = 8$ kV. Turnable tube absent. Critical voltage Φ_{12c} indicated by horizontal lines.

Φ_{12} between the electrodes behaves in a regular way as shown in fig. 3a. However, if the current is increased by raising the initial voltage Φ_{b2}, the plasma voltage reaches a critical value Φ_{12c}, beyond which further acceleration becomes impossible. At the same time irregularities arise as shown in fig. 3b, and the impedance of the discharge decreases. A further enhancement of the bank voltage does not change the critical voltage as long as the magnetic field is kept constant. This is seen in figs. 3c and 3d, in which the plasma only reaches the critical level at earlier times for higher values of the second bank voltage. When a rapid acceleration takes place, the voltage makes a sharp turn at Φ_{12c} and decreases in a highly irrepro-

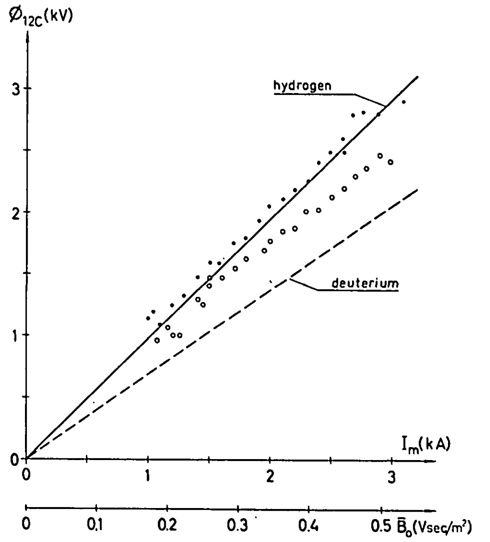

Fig. 4 Critical voltage for hydrogen (dots) and deuterium (rings) as a function of the magnetic field strength. $p_0 = 32$ mTorr. The theoretical results are given by the lines.

ducible way. Thus, the critical voltage manifests itself in a very clear way in the present device.

The variation of Φ_{12c} with the magnetic field strength is demonstrated by fig. 4 for hydrogen and deuterium. The relationship is nearly linear. Measurements with deuterium, and also with heavier gases like helium, yield lower values than those for hydrogen but only after a considerable number of discharges. Further discussions on this fact are postponed to sect. 4C.

It should finally be pointed out that marks due to the interaction with the discharge can be found on the surface of the insulator in fig. 1. They exist only within a zone defined by the field lines which pass between r_{01} and r_{02}, and they are only observed at the side facing the "wind" of the rotating plasma. The damage of the surface is much less than that observed on the Pyrex glass insulators of the earlier device FI.

3B Variable Radial Ratio

A series of measurements have been made with the turnable tube inserted into the apparatus of fig. 1. When the radial ratio is being changed by turning the tube at different angles θ with respect to the equatorial plane, the result becomes as shown in fig. 5. The power input has been chosen high enough for the plasma to reach the highest existing critical value Φ_{12c}, that is when the tube is in the uppermost position and $\theta = 139°$ (fig. 5a). When all other parameters are kept constant and θ is decreasing, the corresponding critical value decreases as shown in figs. 5b–d.

Further results have been obtained for a stronger magnetic field as demonstrated by fig. 6. The figure shows that the critical voltage increases monotonically with the angle θ and the mean radial ratio

$$\bar{r}_0/\bar{r}_w = (r_{01} + r_{02})/(r_{w1} + r_{w2})$$

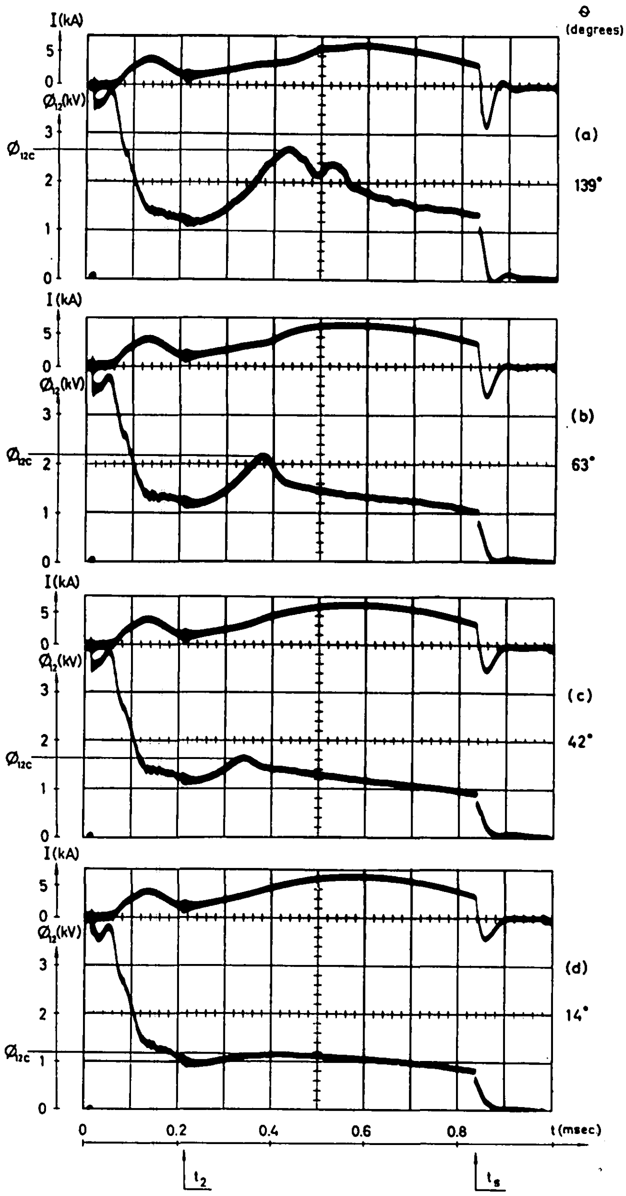

Fig. 5 Bahaviour of plasma current and voltage when the angle θ of the turnable tube is being reduced. The values of θ are: (a) 139,° (b) 63°; (c) 42°; (d) 14°. $I_m = 2.5$ kA, $p_0 = 34$ mTorr., $\Phi_{b1} = 8$ kV, $\Phi_{b2} = 14$ kV. Critical voltages Φ_{12c} indicated by horizontal lines.

To avoid contributions to the plasma voltage from the series "impedance", the measurements have been carried out at the strongest possible magnetic field. This point is further discussed in sect. 4A and in the Appendix.

3C The Effect of Disturbances in the Equatorial Plane

In positions close to the equatorial plane, the turnable tube has a very strong influence on the voltage levels and the plasma impedance. As a further step one should therefore investigate how the voltage and the energy balance depend on the size of an inserted obstacle. For this purpose, movable tubes of different

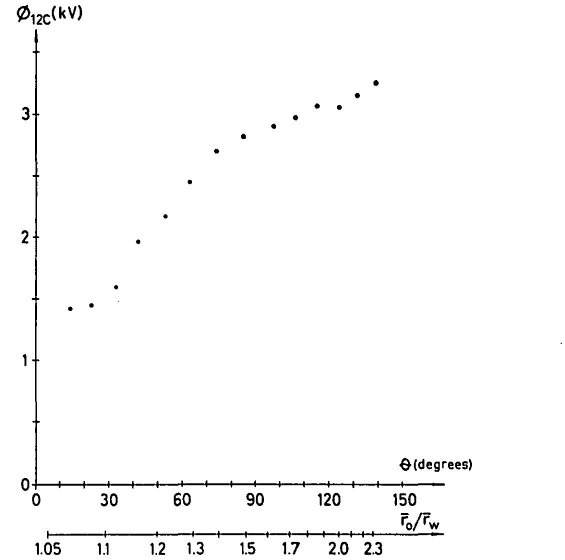

Fig. 6 Critical voltage Φ_{12c} as a function of the angle θ of the turnable tube and the corresponding mean radial ratio $\bar{r}_0/\bar{r}_w$. $I_m = 3.4$ kA, $p_0 = 33$ mTorr., $\Phi_{b1} = 8$ kV, $\Phi_{b2} = 14$ kV.

diameters have been introduced into the equatorial plane and across the entire confinement region, as described in sect. 2 and shown in fig. 1.

The results are shown in fig. 7. Throughout these measurements the second acceleration has been adjusted below the critical voltage Φ_{12c} which exists when there is no tube in the equatorial plane (fig. 7a). With an inserted tube that is only 1.6 mm in diameter, there is a decrease of the voltage by as much as a factor two (fig. 7b). For tubes of increasing diameters there are further reductions in the voltage (figs. 7c and 7d).

The results show that the rotating plasma is heavily disturbed even by very small obstacles in the equatorial plane. This can also be taken as an indication that the losses of the system must be quite small under normal operation.

4 Discussion

Even if the obtained results clearly show that there exists a critical voltage, it is so far not quite obvious how this is connected with the plasma velocity and its distribution in space. We now discuss some of the problems which arise in this connection.

4A The Series Voltage Drop

The equivalent electric circuit of a fully ionized rotating plasma consists of a rather complex system which has been discussed in some detail in earlier papers [22, 23, 15] as well as in the Appendix of this report. In a simplified form the circuit is given within the shaded area of fig. 2. The plasma voltage Φ_{12} is divided into one contribution Φ_ε across the equivalent capacity C and its leakage resistance R due to transverse and longitudinal momentum losses, and into a series voltage drop Φ_{12}' across a kind of series "impedance" which represents the Hall, inertia, and pressure effects arising in Ohm's generalized law.

The plasma voltage Φ_{12} will become a measure of the velocity field v_φ only if Φ_{12}'/Φ_s is a small quantity. This ratio can be estimated as shown in the Appendix. As found from the discussion on eq. 9, the voltage

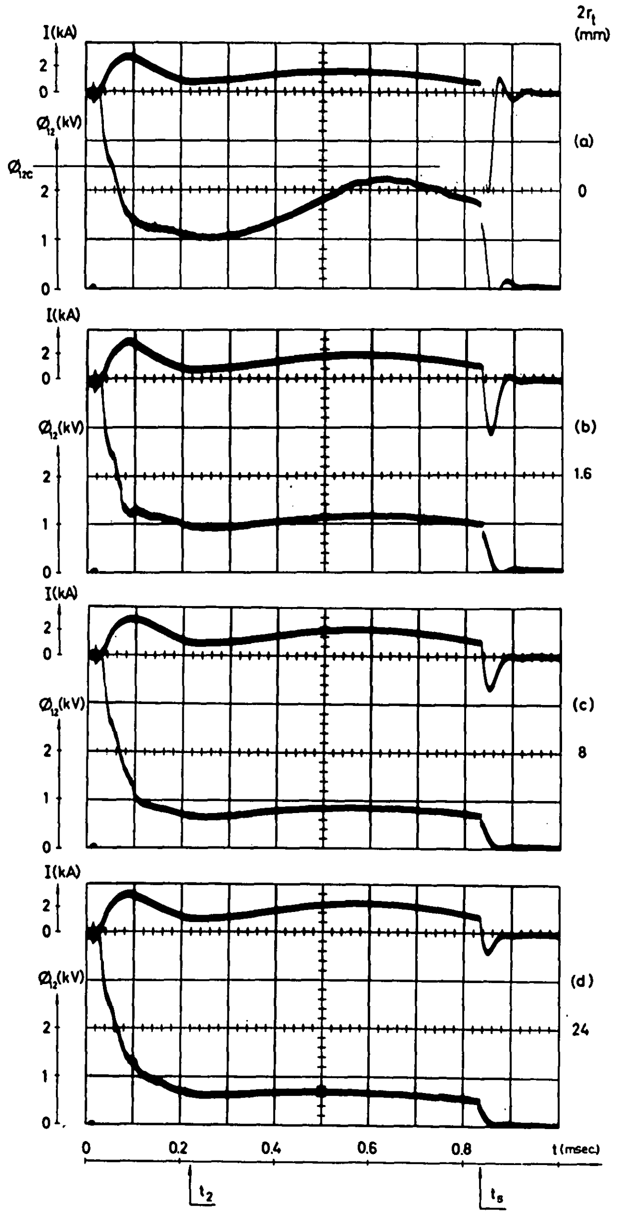

Fig. 7 Behaviour of the discharge when ceramic tubes are inserted radially into the equatorial plane. The diameters of the tubes are: (a) 0 (no tube); (b) 1.6 mm; (c) 8 mm; (d) 24 mm. $I_m = 2.45$ kA, $p_0 = 34$ mTorr., $\Phi_{b1} = 8$ kV, $\Phi_{b2} = 6$ kV. Turnable tube absent.

Φ_{12}' becomes important at high discharge currents, large momentum losses and weak magnetic fields. It will have an influence both in earlier experiments with the Homopolar devices and under special conditions in the present experiments. This is the case when the turnable tube is near the equatorial plane and when the magnetic field is weak.

4B The Velocity Distribution

When the series voltage drop Φ_{12}' can be neglected, the voltage Φ_{12} will be given by the integral of $\mathbf{v}\times\mathbf{B}$ between the electrodes. Since the magnetic field lines can be considered as equipotential surfaces [22] the integration is carried out in the equatorial plane

$$\Phi_{12} = \int_{r_{01}}^{r_{02}} v_\varphi B \, dr = v_0 \int_{r_{01}}^{r_{02}} V B \, dr \tag{1}$$

Here $V = v_\varphi/v_0$ is a normalized expression for the velocity field and v_0 the corresponding characteristic velocity.

The plasma voltage Φ_{12} and its critical value Φ_{12c} depend both on the form of the radial distribution $V(r)$ and on the magnitude v_0. When considering $V(r)$ we have to observe that the experiments described by fig. 4 have been performed at intermediate field strengths. The conditions are then very much the same as in device FI for fields in a range where the transverse momentum losses are larger than the longitudinal [15]. As a first approximation we therefore choose an earlier solution for V which is centered around $r = \frac{1}{2}(r_{01}+r_{02})$ and which drops to zero at $r = r_{01}$ and r_{02} under the influence of the transverse viscosity and heat conductivity [17]. Further, we determine the critical value v_{0c} of v_0 by assuming that a strong insulator interaction sets in as soon as the critical velocity v_c is reached by the plasma somewhere at the insulator surface. This yields

$$v_{0c} = v_c/(r_w V/r_0)_{max} \tag{2}$$

as calculated from $V = V(r_0)$ and the field given in fig. 1. A slight increase beyond the voltage given by eqs. 1 and 2 may become possible if the velocity at the insulator is allowed to approach v_c along a small part of the surface, and not only at one point. On the other hand, if v_φ approaches v_c along the entire surface, the voltage would increase gradually up to a saturation level about twice as high as that given here, but this can hardly be reconciled with the sharp onset of the critical voltage observed in the experiments.

The results obtained for hydrogen and deuterium from eqs. 1 and 2 are shown in fig. 4. The theory agrees with the experiments with hydrogen, and even better than what can be expected from the approximations involved. That the measured values for deuterium differ by less than a factor of $\sqrt{2}$ from those for hydrogen will be further discussed in sect. 4C.

In the experiments of fig. 6 with the turnable tube, we should expect the momentum and energy balances to change rather much with the angle θ. As a consequence, not only the magnitude of v_{0c} but also the shape of the velocity distribution V will vary with θ, and the simple solution for v_φ earlier used in this section is no longer valid within the whole range of θ. This affects the integral of eq. 1 in a rather complicated way. At the present stage we can only state that Φ_{12c} is observed to increase by a factor of 2.3 when the mean radial ratio r_0/r_w is increased by 2.2. This is roughly what should be expected.

4C Critical Voltage in Gases Heavier than Hydrogen

In several experiments the critical voltage in heavier gases is higher than what should be expected from a comparison with the results for hydrogen. A similar observation has earlier been reported by Angerth et al. [4] who obtained two voltage plateaus for heavier gases with a system having unbaked electrodes and a mirror ratio of 1.2. The first plateau appeared after the start of the discharge and was consistent with the gas being used. After a short time a higher plateau was observed, somewhere between the level of the first and that in a discharge with pure hydrogen. The second plateau could be removed by baking the electrodes. The same authors suggested that the second plateau was due to hydrogen impurities, and that the heavier gas was centrifuged out.

A possible explanation of these results as well as of those present in sect. 3A can be given for systems where the radial ratio exceeds unity [24, 25, 26, 22, 16, 27]. At a given temperature and velocity of rotation and under the influence of the centrifugal force, the density gradient of the plasma along the field lines becomes much steeper for heavy than for light gases [24, 26, 27]. As a consequence, the gas close to the end insulators should consist mainly of released hydrogen, even if there is only a small fraction of hydrogen within the main parts of the plasma. Since the critical voltage depends on a plasma-neutral interaction in a thin layer at the insulator surface, the velocity limit will be determined by the hydrogen gas, and not by the heavier gases. A hydrogen gas layer of this type should not arise in the homogeneous magnetic field of a Homopolar device having a radial ratio of unity.

The effect discussed here can also be used to raise the accessible critical voltages of heavier gases by mixing a certain amount of hydrogen into the discharge.

4D The Stability Problem

The present FII device appears to have the same stability properties and the same type of particle, momentum, and energy balance as the earlier FI device [15], where there is so far no trace of flute phenomena. In FII, field lines pass freely around the main coil for about 88% of the perimeter. Thus, the fact that the main part of the plasma does no longer "lean" against an end insulator and that the spoke-shaped rod is asymmetric do not seem to affect the plasma behaviour as far as flutes are concerned.

A full explanation of the critical voltage has not yet been given. From the present results, it can at least be seen that it is not due to such instabilities which set in at a certain value of the ratio $\beta_c = \mu_0 nm\, v_\varphi^2/B^2$ between the energy densities of the fluid motion and the magnetic field. With v_φ proportional to v_c, as observed in the experiments, we should have β_c inversely proportional to B^2 and not equal to a constant.

5 Conclusions

(i) In a rotating plasma there exists a sharply defined critical voltage and a corresponding velocity limit. An increasing power input does not raise the plasma velocity beyond this limit, but increases the losses and changes the behaviour of the discharge.

(ii) This phenomenon appears in a very clear way in the present device, where the plasma voltage slowly approaches the critical value from below and where the insulator surface consists only of a narrow rod.

(iii) There are strong indications that the critical voltage is due to an interaction between the plasma and a thin neutral gas layer close to the insulator surface. This should take place as soon as the plasma velocity reaches v_c within the layer. Thus, the voltage limitation occurs in a region which is very small compared to the plasma volume and which is clearly defined by the magnetic field lines, the positions of the electrodes and the insulator surface.

(iv) When full use is being made of the radial ratio, a critical voltage is reached which agrees well with that calculated from the velocity v_c, the radial ratio r_0/r_w, and the shape of the velocity distribution. The latter is deduced from the momentum and energy balance. The highest accessible velocity should then become about $(r_0/r_w)v_c$.

(v) The critical voltage increases with a variable radial ratio, roughly in a way excepted from theory.

(vi) The plasma is strongly disturbed, even by a very small obstacle which is placed in the equatorial plane. This can be taken as an indication that the losses of the system must be small under normal operation.

(vii) At least with respect to flutes, the stability is not influenced by the fact that the field lines pass freely around the main coil for about 88% of the perimeter. The critical voltage is not associated with such instabilities which depend on a critical ratio between the centrifugal and magnetic "pressures".

Appendix

B. Lehnert

The Equivalent Circuit

We consider a quasi-neutral fully ionized rotating plasma in the long-cylinder approximation, where there is a homogeneous magnetic field $\mathbf{B}$ along the axis of rotation and where the momentum balance can be expressed by

$$n\,m\,\frac{d\mathbf{v}}{dt} = n\,m\left(\frac{\partial}{\partial t} + \mathbf{v}\cdot\nabla\right)\mathbf{v} = \mathbf{i}\times\mathbf{B} - \nabla p + \mathbf{F} \tag{3}$$

and

$$\eta\,\mathbf{i} = \mathbf{E} + \mathbf{v}\times\mathbf{B} - \frac{m_i - m_e}{e\,n\,m}\,\mathbf{i}\times\mathbf{B} + \frac{1}{e\,n\,m}\,(m_i\,\nabla p_e - m_e\,\nabla p_i) \tag{4}$$

In these equations n denotes the particle density, $\mathbf{v}$ is the plasma velocity, $\mathbf{E}$ is the electric field, $\mathbf{i}$ is the current density, $m=m_i+m_e$, $p=p_i+p_e$, p_i and p_e are the ion and electron pressures, $\eta=m_e\nu_{ei}/e^2n$ is the resistivity, and ν_{ei} is the electron-ion collision frequency. In a cylindrical frame of reference (r, φ, z), the momentum losses are represented by the force

$$\mathbf{F}=-\mu^*\nabla\times\nabla\times\mathbf{v}-n\,m\,\nu_{\|}\,\mathbf{v}_\varphi\equiv\mathbf{F}_\mu+\mathbf{F}_{\mathrm{L}} \quad (5)$$

where

$$\mu^*=p_i/\nu_{ii}(1+4\,\omega_i^2/\nu_{ii}^2) \quad (6)$$

is the transverse viscosity due to ion-ion collisions which take place at the frequency ν_{ii}, and $\omega_i=eB/m_i$. Further, $\nu_{\|}$ is the effective frequency at which momentum is being lost along the field lines to the end insulators of the plasma device. Thus, the transverse and longitudinal losses of momentum are represented by the forces $\mathbf{F}_\mu$ and $\mathbf{F}_{\mathrm{L}}$, which are both in the φ direction. In this long cylinder approximation we consider $\mathbf{F}_{\mathrm{L}}$ as an equivalent force applied inside the whole volume of the plasma.

In terms of components, eq. 3 becomes

$$i_r=-\frac{n\,m}{B}\left(\frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t}\right)_\varphi+\frac{1}{B}(F_\mu+F_{\mathrm{L}})\equiv i_{r\varepsilon}+i_{rf} \quad (7)$$

$$i_\varphi=\frac{n\,m}{B}\left(\frac{\mathrm{d}\mathbf{v}}{dt}\right)_r+\frac{1}{B}\frac{\partial p}{\partial r}\equiv i_{\varphi\varepsilon}+i_{\varphi p} \quad (8)$$

Equations 3, 4, 7, 8 can further be combined to

$$E_r/B=-v_\varphi+\frac{\nu_{ei}\,F}{n\,m\,\omega_i\,\omega_e}+\frac{1}{\omega_i}\left(\frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t}\right)_r-\frac{\nu_{ei}}{\omega_e\,\omega_i}\left(\frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t}\right)_\varphi$$
$$+\frac{1}{n\,m\,\omega_i}\left(\frac{\partial p_i}{\partial r}-\frac{m_e}{m_i}\frac{\partial p_e}{\partial r}\right)\equiv-v_\varphi+E_r'/B \quad (9)$$

$$E_\varphi/B=v_r-\frac{F}{n\,m\,\omega_i}+\frac{1}{\omega_i}\left(\frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t}\right)_\varphi+\frac{\nu_{ei}}{\omega_e\,\omega_i}\left(\frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t}\right)_r$$
$$+\frac{\nu_{ei}}{n\,m\,\omega_i\,\omega_e}\frac{\partial p}{\partial r}\equiv v_r+E_\varphi'/B \quad (10)$$

When magnetic induction effects are neglected, we have $E_\varphi=0$.

The equivalent electric circuit can easily be deduced from the present equations (see also refs. 22, 23, 15). Equation 7 shows that the radial current is divided into one part $i_{r\varepsilon}$ which passes through a capacity corresponding to the equivalent dielectric constant nm/B^2, and into one part i_{rf} which is consumed by frictional forces due to the transverse and longitudinal momentum losses. From eq. 9 it is seen that the radial electric field is divided into one part corresponding to the velocity v_φ and its associated voltage across the equivalent capacity, and into one part E_r' which includes Hall, inertia, and pressure effects. A similar interpretation can be given to eqs. 8, 10 for the radial velocity v_r of diffusion and the azimuthal current i_φ which is coupled to the radial current by the Hall effect. In eq. 8 the current $i_{\varphi\varepsilon}$ arises from the centrifugal force as well as from a small accumulation of kinetic energy by the radial motion, and $i_{\varphi p}$ is the diamagnetic current balancing the pressure gradient.

When this long-cylinder approximation is applied to the configuration of fig. 1 and the equations are integrated over the space between the electrodes, an equivalent circuit is obtained which has the form given within the shaded area of fig. 2. Thus, the plasma voltage Φ_{12} is divided into one part Φ_ε across the equivalent capacity C and its leakage resistance R due to the frictional forces, and one part Φ'_{12} which includes both potential drops and electromotive forces due to Hall, inertia, and pressure effects. A similar result can be deduced for an arbitrary cylindrical geometry and for a partially ionized plasma.

The magnitude $\Phi_{12}'/\Phi_\varepsilon$ is easily estimated by means of eqs. 9, 5, 6. It becomes small when ω_i is large compared to ν_{ei}, ν_{ii}, $\nu_{\|}$ and v_φ/r and when the ion Larmor radius is small compared to the characteristic macroscopic dimensions.

As an illustration we choose the data of the present experiments with $n\sim2\times10^{21}\,\mathrm{m}^{-3}$, $v_\varphi\sim10^5\,\mathrm{m\,sec}^{-1}$, ion and electron temperatures in the range from 10^5 to 5×10^5 °K, a characteristic time scale of some 100 microseconds, and linear dimensions and magnetic field strengths given by figs. 1, 3–7. Then, the magnitude of the second and fourth terms in the intermediate member of eq. 9 should be less than 0.03 of the first term. The corresponding magnitudes of the third and fifth terms should exceed 0.1 at magnetic field strengths below 0.2 $V\,\mathrm{sec\,m}^{-2}$, as well as in a strongly dissipative system where the effective frequency $\nu_{\|}$ of momentum losses exceeds $10^6\,\mathrm{sec}^{-1}$ considerably. This shows that the series voltage drop Φ_{12}' cannot always be neglected.

Acknowledgments

The authors are indebted to Dr. B. Bonnevier for checking the magnetic field on a computer, and to Dr. U. Fahleson for valuable discussions. The authors also express their thanks to Mr. R. Ekman, Mr. E. Christoffersen, Mr. T. Wellroth, and Mr. H. Sannemark for valuable help with the construction and building of the experimental equipment. The important assistance of Mr. L. Hagman, Mr. D. Haslbrunner, Mr. G. Hägerström, Mr. G. Kindberg, Mr. L. Jonsson, and Mr. K. Ohlén is also gratefully acknowledged.

This work has been supported financially by the Swedish Atomic Research Council, the Research Institute of National Defence, the Swedish Atomic Energy Company, and the Royal Institute of Technology.

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(Manuscript received 24 February 1966)